

Lumina-SIS

Multi-Tenant School Information System
Complete Technical & Functional Documentation

Version 1.0 · Generated 29 August 2026

Table of Contents

1. Introduction & Overview
2. System Architecture
3. Technology Stack
4. Multi-Tenancy & Data Isolation
5. Roles, Permissions & Security
6. Core Modules & Features
7. Academics: Grading, Gradebook & Attendance
8. Report Card System & Template Designer
9. MHPS Upper School Report Card (Tenant Add-on)
10. Student & Staff Profiles
11. Data Model (Collections)
12. API Reference
13. Integrations
14. Deployment & Environment
15. Glossary

1. Introduction & Overview

Lumina-SIS is a modern, multi-tenant School Information System that lets one deployment serve many independent schools ('tenants') from a single application and database, with strict data isolation between them. It centralises the day-to-day administration of a school: admissions, student and staff records, class organisation, attendance, gradebook and grading, discipline and health records, and the generation of professional term report cards.

What problem it solves

- Replaces paper registers, spreadsheets and disconnected tools with one source of truth.
- Gives each school its own branded environment (logo, colours, subjects, grading) while sharing one platform.
- Automates report card production — from teacher score entry to a print-ready, tenant-branded PDF.
- Enforces who can see and do what, through role-based access control (RBAC).

Who uses it

Superuser	Platform operator	Creates schools, manages all tenants, designs report templates, full override access.
Admin	School administrator	Manages the school's students, staff, classes, admissions, settings and report configuration.
Teacher	Class / form teacher	Enters attendance, gradebook scores, comments, behaviour, generates reports for their classes.
Parent	Guardian	Views their child's finalised report cards and information (read-only).

2. System Architecture

Lumina-SIS follows a classic three-tier architecture with a clear separation between the presentation layer (React SPA), the application layer (FastAPI REST API) and the data layer (MongoDB).

High-level flow

Frontend (SPA)	React 19 + Tailwind CSS	UI, routing, forms, charts, client-side PDF generation.
API Gateway	Kubernetes Ingress	Routes '/api/*' to the backend (port 8001) and all other paths to the frontend (port 3000).
Backend (API)	FastAPI (Python)	Business logic, authentication, RBAC, validation, persistence.
Database	MongoDB (Motor async driver)	Document storage; all records keyed by UUID and scoped by school_code.
Process manager	Supervisor	Runs & restarts the frontend and backend services.

Key design principles

- **API prefix:** every backend route is served under `/api` to satisfy the ingress routing rules.
- **UUID identifiers:** all documents use string UUIDs (never Mongo ObjectIDs) so payloads are JSON-serialisable and portable.
- **Environment-driven config:** the frontend calls the backend via `REACT_APP_BACKEND_URL`; the backend reads `MONGO_URL / DB_NAME` — no hardcoded URLs.
- **Stateless auth:** JWT bearer tokens; no server-side session store.

3. Technology Stack

Backend

FastAPI	0.110.1	Web framework / REST API
Uvicorn	0.25.0	ASGI server
Motor / PyMongo	3.3.1 / 4.5.0	Async MongoDB driver
Pydantic	2.12.5	Request/response validation & models
PyJWT	2.10.1	JWT access-token signing/verification
bcrypt / passlib	4.0.1 / 1.7.4	Password hashing
pandas / openpyxl	2.3.3	CSV/Excel import & export
resend	>=2.0.0	Transactional email (report card delivery, password reset)

Frontend

React	19.0	UI library
react-router-dom	7.5	Client-side routing
Tailwind CSS	3.x	Utility-first styling + design tokens
axios	1.8	HTTP client
recharts	3.6	Dashboard & attendance charts
html2canvas + jsPDF	1.4 / 4.2	Client-side report card PDF export
lucide-react	0.507	Icon set
sonner	2.0	Toast notifications
Radix UI	1.x	Accessible primitives (dialog, select, tabs, etc.)

4. Multi-Tenancy & Data Isolation

A single Lumina-SIS deployment hosts many schools. Every school is identified by a unique, uppercase **school_code** (e.g. MHPS, JTECH). Virtually every document — students, users, classes, attendance, gradebook, report templates, comment banks — carries a `school_code` field.

How isolation is enforced

- Every authenticated request carries the user's `school_code` inside the JWT.
- Queries are automatically filtered by that `school_code`, so a user of one school can never read or write another school's data.
- **Superuser bypass:** superusers are platform-level and can view/act across all tenants (used for support, template design and impersonation).
- Tenant-locked features (e.g. the MHPS report card) additionally assert the caller's `school_code` matches the owning tenant.

Per-tenant configuration

- Branding: logo/crest, primary & accent colours, motto/tagline, address, principal signature.
- Academics: custom subject list, houses, academic years (with a 'current' year and enable/disable toggles).
- Reporting: its own report template, grade scale labels and, where applicable, a school-specific comment bank.

5. Roles, Permissions & Security

Authentication

- Login requires **school_code + username + password** (a superuser logs in against their platform school).
- Passwords are hashed with **bcrypt**; plaintext is never stored.
- On success the API returns a signed **JWT access token**; the frontend attaches it as a Bearer token on every request.
- Password recovery is available via `/api/auth/forgot-password` and `/api/auth/reset-password` (email-based).

The four roles

Access is governed by role plus a set of granular permissions. The eight permission flags are:

manage_schools	Create/edit schools, academic years, subjects, branding.
manage_users	Create/edit staff & parent accounts, assign roles.
manage_students	Create/edit student records, profiles, enrolment.
manage_classes	Create/edit classes and class assignments.
manage_attendance	Record and edit attendance.
manage_grades	Enter and edit gradebook scores & comments.
view_reports	View report cards and analytics.
generate_reports	Produce & send finalised report cards.

Role capability matrix

Manage schools / tenants	Yes	Own school	No	No
Manage users & roles	Yes	Yes	No	No
Manage students & classes	Yes	Yes	Assigned	No
Enter attendance / grades	Yes	Yes	Assigned classes	No
Design report templates	Yes	No	No	No
Configure report content*	Yes	Yes	No	No
Generate / send reports	Yes	Yes	Own classes	No
View finalised report cards	Yes	Yes	Yes	Own child
Audit log access	Yes	Yes	No	No

*e.g. grade-scale labels, comment bank wording, signature blocks. Layout/structure remains Superuser-only.

Auditability

Sensitive actions (create, update, delete, clone, lock/unlock, sending reports) are written to an **audit log** capturing the actor, action, entity and timestamp, viewable by admins and superusers.

6. Core Modules & Features

The application is organised into functional areas surfaced through the left navigation. Each maps to one or more pages and API groups.

Overview / Dashboard

At-a-glance KPIs — student & staff counts, attendance summaries, and quick links tailored to the user's role.

Admissions

Capture inquiries and applications, track their stage, and convert accepted applicants into enrolled students in one click.

Re-Enrollment

Bulk promote/roll students into the next academic year with a preview-then-execute workflow (enrolment runs).

People — Students

Continuous student profile: identity, contact, academics, attendance, behaviour, family, medical, report fields and school info.

People — Staff

Staff profile: identity, contact, role & access, and login-management (credential/role changes).

People — Users & Roles

Create staff/parent accounts, set roles and per-user permissions, reset credentials.

Academics — Classes

Define classes/forms, grade levels, assign a form teacher and subjects.

Academics — Attendance

Daily attendance capture (single & bulk) with present/absent/tardy/excused states and per-student summaries.

Academics — Gradebook

Per-class, per-subject score entry with grade distribution analytics and record locking.

Report Manager

Class lists, gradebook access, social-skills entry, form-teacher comments and term report generation.

Student Services — Health

Allergies, chronic conditions, medications, vaccinations and clinic visit logs per student.

Student Services — Discipline

Log discipline incidents with categories/severity and view discipline statistics.

Administration — Schools

Create/edit schools, branding, academic years, subjects and signatures.

Administration — Import / Export

Bulk import students & teachers from CSV templates; export data templates.

Administration — Report Designer

Three-zone (header/body/footer) template designer with a system-wide default and per-school overrides.

Administration — Audit Log

Immutable trail of key actions across the tenant.

7. Academics: Grading, Gradebook & Attendance

Default grading scheme

Numeric scores map to letter grades and performance domains. Each school can further customise its own scale where a template supports it.

Score Range	Letter Grade	Performance Domain	Score
90–100	A+	Expert performance	4.0
85–89	A	Highly Proficient	3.8
80–84	A-	Proficient	3.7
75–79	B	Satisfactory	3.5
70–74	B-	Developing	3.3
65–69	C	Passing	3.2
60–64	C-	Passing	2.8
55–59	D	Marginal	2.6
50–54	D-	Below Average	2.4
40–49	E	Frustration	1.0
0–39	U	No participation	0

Gradebook

- Scores are entered per class → subject → student → term.
- A **grade distribution** endpoint powers histogram analytics for a class.
- Records can be **locked/unlocked** to freeze grades once finalised.

Attendance

- States: **Present, Absent, Tardy, Excused**.
- Single-entry and **bulk** capture for a whole class/day.
- Per-student attendance **summaries** feed the profile donut chart and report cards (days in term / days absent).

8. Report Card System & Template Designer

Report cards are produced from a **template** plus each student's term data. Lumina-SIS ships a system-wide default template (the 'Ashcombe Academy' design) and lets schools override it.

Template architecture

- **System default:** a global template (scope = 'global') used by every school that hasn't customised its own.
- **Per-school templates:** stored keyed by `school_code`; a school can clone the system default and adjust header, body and footer zones, theme colours and signatures.
- **Three zones:** Header, Body and Footer are configured independently (endpoints exist for each zone plus theme).
- Superusers design structure/layout; schools configure content and branding.

Generation & delivery

- Teachers enter scores, comments, social skills and behaviour; the term report is assembled per student.
- Report cards can be **locked** (per student or per class) to prevent further edits once issued.
- Finalised cards are rendered to PDF client-side (html2canvas + jsPDF) and can be **emailed** to guardians via Resend.
- Parents access finalised PDFs read-only through their portal.

9. MHPS Upper School Report Card (Tenant Add-on)

A second, **tenant-locked** template built specifically for **Mona Heights Primary School** (school_code MHPS), Grades 4–6. It is only selectable, editable and renderable for MHPS (superusers bypass), and it never becomes the platform default — every other school keeps the Ashcombe default untouched.

What it adds

- **Extended profile fields:** house, reading level, post of special responsibility, extra-curricular activities.
- **Report record** per student/term: position in class, number in class, days in term/absent, term & period labels, and an auto-calculated overall average (with manual override).
- **Six assessment components** per subject (class tests/quizzes, collaborative task, homework, projects, mid-term, end-of-year) with an auto-computed, weight-based **Weighted Term Grade** and derived letter.
- **Achievement Standards** module: core-subject percentages auto-mapped to 4 proficiency bands (Highly Proficient / Proficient / Developing / Beginning) with editable descriptions.
- **Performance Task** and **Behaviour/Conduct ratings** (Work & Personal Ethics + Social Skills, EX/VG/G/NI scale).
- **Comment bank:** 13 seeded stock comments, school-Admin editable; teachers multi-select + free-text additional comments.
- **Configurable settings:** academic grade scale, achievement bands, behaviour criteria, houses and principal signature block.

Output

Rendered to **US Legal (8.5in × 14in)** PDF, matching the school's paper form, with the green/gold MHPS branding pulled from tenant settings. RBAC: Superuser = structure, Admin = configurable content, Teacher = data entry, Parent = view PDF.

10. Student & Staff Profiles

Both profiles use a single continuous, scrollable layout with a shared visual language.

- **Status ribbon** header: name left, status pill right, and key facts (grade, ID, class, year / username, email, phone) inline underneath.
- **Sticky pill anchor chips** below the header jump to each section and highlight as you scroll.
- **Sectioned content:** Students — Dashboard, Student Info, Contact, Academics, Attendance, Behavior, Family, Medical, Report Fields, School. Staff — Dashboard, Profile, Contact, Login Management.
- **Accent cards:** white surface with a 3px left accent border — blue for identity/contact/school data, navy for academic/attendance/security data.
- **Collapsible roster rail** with a toggle; the active row is marked by a left-edge blue indicator bar.
- Attendance donut and charts retain their green/amber/red status colours.

11. Data Model (Collections)

All collections use string UUID ids and (where tenant-owned) a school_code field.

schools	Tenant records: code, name, branding, academic years, subjects, principal.
users	Staff & parent accounts: role, permissions, hashed password, profile.
students	Student master records & profile (incl. MHPS extended fields).
classes	Classes/forms: grade level, form teacher, subjects.
attendance	Daily attendance entries per student.
gradebook	Per class/subject/student/term scores & lock state.
report_templates	System default + per-school templates (header/body/footer, theme, MHPS settings).
report_cards	MHPS per-student/term report records (components, achievement, behaviour, comments).
report_card_locks	Lock state for finalised report cards.
comment_bank	Tenant-scoped pre-written teacher comments.
teacher_comments	Per-student form-teacher comments.
social_skills	Social-skills/attitude ratings per student.
discipline_incidents	Behaviour/discipline log.
health_records	Allergies, conditions, medications, vaccinations, visits.
admissions	Inquiries & applications and their stage.
enrollment_runs	Re-enrolment / promotion batches.
signatures	Uploaded signature images for reports.
audit_logs	Immutable action trail.
password_resets	Password reset tokens.

12. API Reference

The REST API exposes ~123 endpoints, all under the `/api` prefix and (except auth) requiring a Bearer token. Grouped by area:

Authentication

POST	/auth/login	Log in (school_code + username + password) → JWT
GET	/auth/me	Current authenticated user
POST	/auth/forgot-password	Request a password-reset email
POST	/auth/reset-password	Complete a password reset

Schools & Config

GET/POST	/schools	List / create schools
GET/PUT/DELETE	/schools/{id}	Read / update / delete a school
POST/PUT/DELETE	/schools/{id}/academic-years[...]	Add, rename, toggle, set-current, delete a year
GET/POST/PUT/DELETE	/schools/{id}/subjects[...]	Manage a school's subject list
POST	/schools/{id}/signatures/upload	Upload a signature image
GET	/system/context	Current academic year & enabled list
GET	/grading-scheme, /subjects, /houses, /permissions	Reference lookups

Users & Roles

GET/POST	/users	List / create staff & parent accounts
GET/PUT/DELETE	/users/{id}	Read / update / delete a user
PUT	/users/{id}/role	Change a user's role
PUT	/users/{id}/credentials	Reset username/password
GET	/teachers, /parents	Role-filtered user lists

Students

GET/POST	/students	List / create students
GET/PUT/DELETE	/students/{id}	Read / update / delete a student

PUT	/students/{id}/report-fields	Update report-specific fields
GET	/students/{id}/attendance/summary	Attendance summary

Classes & Attendance

GET/POST	/classes	List / create classes
GET/PUT/DELETE	/classes/{id}	Read / update / delete a class
GET/POST	/attendance	Query / record attendance
POST	/attendance/bulk	Bulk attendance capture

Gradebook

GET/POST	/gradebook	Query / enter scores
DELETE	/gradebook/{id}	Remove a gradebook entry
POST	/gradebook/{id}/lock unlock	Lock / unlock a record
GET	/gradebook/{class_id}/distribution	Grade distribution analytics

Report Templates

GET	/report-templates/system-default	Fetch the global default
GET/PUT	/report-templates/{code}	Read / update a school's template
PUT	/report-templates/{code}/header body footer theme	Update a zone / theme
POST	/report-templates/{code}/clone-system-default	Clone the default for a school
POST	/report-templates/{code}/reset-default	Reset a school to the default

Report Cards

GET	/report-card/{student_id}	Assembled report payload
GET	/report-cards/class/{class_id}	Class batch
POST/DELETE	/report-cards/{student_id}/lock	Lock / unlock a student's card
GET	/report-cards/locks	Lock states
POST	/report-cards/send	Email finalised cards

MHPS (tenant-locked)

GET	/mhps/report-template	MHPS template + settings + comment bank
PUT	/mhps/report-template/settings	Admin: edit configurable content
PUT	/mhps/report-template/structure	Superuser: edit structure
GET/POST/PUT/DELETE	/mhps/comment-bank[...]	Comment bank CRUD
GET/PUT	/mhps/report-card/{student_id}	Read / upsert (auto-calc) a report record

Student Services

GET/POST	/health/{student_id}[...]	Allergies, conditions, medications, vaccinations, visits
GET	/health/stats	Health statistics
GET/POST/PUT/DELETE	/discipline[...]	Discipline incidents CRUD
GET	/discipline/stats	Discipline statistics
GET/POST	/social-skills[...]	Social-skills ratings
GET/POST	/teacher-comments[...]	Form-teacher comments

Admissions & Enrolment

GET/POST	/admissions	List / create admissions
GET	/admissions/inquiries applications stats	Filtered views & stats
POST	/admissions/{id}/convert	Convert applicant → student
GET	/enrollment/preview	Preview a re-enrolment run
POST	/enrollment/execute	Execute a re-enrolment run

Import / Export & Media

POST	/import/students teachers	Bulk CSV import
GET	/export/students-template teachers-template	Download CSV templates
POST	/upload/photo template-background	Upload media
GET	/uploads/{filename}	Serve an uploaded file

Analytics & Audit

GET	/stats/dashboard	Dashboard KPIs
GET	/audit-logs	Audit trail

13. Integrations

Resend (transactional email)

- Used to email finalised report cards to guardians and to send password-reset messages.
- Configured via `RESEND_API_KEY` and `SENDER_EMAIL` environment variables.
- If no key is present, email features are simply inactive — the rest of the system is unaffected.

Client-side PDF

Report cards are captured from the rendered DOM with **html2canvas** and packaged with **jsPDF** — the standard Ashcombe cards on Letter size, the MHPS card on US Legal (8.5in × 14in). The renderers deliberately avoid CSS gradients, box-shadows and backdrop-blur, which do not rasterise reliably in html2canvas.

14. Deployment & Environment

Services

Frontend (React)	3000	Served for all non-/api paths.
Backend (FastAPI)	8001	Bound to 0.0.0.0; all routes under /api.
MongoDB	27017	Local instance via MONGO_URL.

Environment variables

REACT_APP_BACKEND_URL	frontend/.env	Base URL the SPA uses to reach the API.
MONGO_URL	backend/.env	MongoDB connection string.
DB_NAME	backend/.env	Database name.
JWT_SECRET	backend/.env	Secret used to sign access tokens.
CORS_ORIGINS	backend/.env	Allowed CORS origins.
RESEND_API_KEY / SENDER_EMAIL	backend/.env	Email delivery (optional).

Operations

- Both services run under **Supervisor** (`supervisorctl restart backend|frontend|all`) with hot-reload in development.
- Seed scripts under `/app/scripts` provision demo tenants, students, gradebook and the MHPS template.
- A superuser account is auto-created on backend startup if none exists.

15. Glossary

Tenant / School. An independent school served by the shared platform, identified by `school_code`.

Superuser. Platform-level operator with cross-tenant access.

RBAC. Role-Based Access Control — permissions derived from a user's role.

JWT. JSON Web Token — the signed bearer token used for authentication.

Report template. The reusable design (header/body/footer + theme) a report card is built from.

Weighted Term Grade. MHPS: a subject grade computed from six weighted assessment components.

Achievement Standards. MHPS: proficiency bands (Highly Proficient → Beginning) for the four core subjects.

Comment bank. A tenant-scoped library of pre-written teacher comments.

Enrolment run. A batch operation that promotes/rolls students into a new academic year.

Lock. Freezing a gradebook entry or report card to prevent further edits.

End of document · Lumina-SIS Documentation